1.5.7 Wrap-Up: Ticket Out the Door

1. Find the value of the variable.

   a. $\sqrt{14} = 14^{\frac{1}{k}}$

   $k =$ _____

   b. $50^{\frac{1}{3}} = \sqrt[n]{50}$

   $n =$ _____

2. Complete the table by writing an equivalent expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$6^{-\frac{1}{6}} \cdot 6^{\frac{19}{6}}$		
$\dfrac{7^{\frac{1}{2}}}{7^{\frac{5}{2}}}$		

Unit 1, Lesson 6: Writing Equivalent Expressions Using Rational Exponents

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can apply the Laws of Exponents to rational exponents.
- ☐ I can convert between expressions involving rational exponents and expressions involving radicals.

1.6.1 Warm-Up: Identifying Equivalent Expressions

Three lists of numbers are shown.

List A	List B	List C
$8 \cdot 8 \cdot 8$	2^9	24
$4 \cdot 16 \cdot 2$	$6 \cdot 4$	$\frac{1}{32}$
$2 \cdot 16$	$\left(\frac{1}{2}\right)^5$	512
$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}$	$2^2 \cdot 2^3$	128
$4 + 4 + 4 + 4 + 4 + 4 + 4 + 4$	$2 \cdot 4^3$	32

1. Choose an expression from List A and match it with an equivalent expression from List B and List C.

2. Explain to your partner how you know the expressions you matched are equivalent.

3. Listen to your partner explain the match they found. If you disagree, discuss your thinking and work to reach an agreement.

1.6.2 Guided Practice: Writing Equivalent Expressions

1. Use the power of a power law of exponents to write an equivalent exponential expression for the following two expressions.

   a. Expression 1: $\left(a^{\frac{1}{c}}\right)^b = a^{(\_\_\_\_\cdot\_\_\_\_)} = \_\_\_\_\_\_$

   b. Expression 2: $(a^b)^{\frac{1}{c}} = a^{(\_\_\_\_\cdot\_\_\_\_)} = \_\_\_\_\_\_$

   c. Explain why the expressions $\left(a^{\frac{1}{c}}\right)^b$ and $(a^b)^{\frac{1}{c}}$ are equivalent.

2. Two students evaluated an expression using two different methods.

Jerry	Antonio
$\left((81)^{\frac{1}{2}}\right)^3$	$\left((81)^{\frac{1}{2}}\right)^3$
$(\sqrt{81})^3$	$81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}}$
$(9)^3$	$\sqrt{81} \cdot \sqrt{81} \cdot \sqrt{81}$
729	$9 \cdot 9 \cdot 9$
	729

Explain the difference between the two methods.

1.6.3 Exploration: Matching Expressions

1. Use your calculator to help you match each expression in the first table with an equivalent expression in the second table. Record your answers in the table below.

Rational Exponent
$(64)^{\frac{3}{2}}$
$(64)^{-\frac{1}{2}}$
$(64)^{\frac{2}{3}}$
$(64)^{\frac{1}{4}}$
$(64)^{\frac{1}{2}}$
$(64)^{\frac{4}{3}}$

Radical Expression
$\sqrt[4]{64}$
$\sqrt[3]{64^2}$
$\left(\sqrt[3]{64}\right)^4$
$\sqrt{64}$
$\left(\sqrt{64}\right)^3$
$\frac{1}{\sqrt{64}}$

Rational Exponent	Radical Expression	Rational Exponent	Radical Expression
$(64)^{\frac{3}{2}}$		$(64)^{\frac{1}{4}}$	
$(64)^{-\frac{1}{2}}$		$(64)^{\frac{1}{2}}$	
$(64)^{\frac{2}{3}}$		$(64)^{\frac{4}{3}}$	

2. Discuss with your partner the relationship between the rational exponent and the radical expression.

3. Work with your partner to complete the following statement.

$$\left(a^{\frac{1}{c}}\right)^b = (a^b)^{\frac{1}{c}} = \left(\sqrt[\_\_\_]{a}\right)^{\_\_\_} = \sqrt[\_\_\_]{(a^{\_\_\_})}$$

1.6.4 Activity: Card Sort

1. Work with your partner to sort the cards into sets of equivalent expressions.

$(18^2)^{\frac{1}{3}}$	$\left(21^{\frac{1}{4}}\right)^5$	$18^{\frac{3}{2}}$	$(21^5)^{\frac{1}{4}}$
$\left(\frac{1}{21^4}\right)^{\frac{1}{5}}$	$\frac{1}{(\sqrt[4]{21})^{-5}}$	$21^{\frac{5}{4}}$	$(18^3)^{\frac{1}{2}}$
$\left(18^{\frac{1}{2}}\right)^3$	$\left(21^{\frac{1}{5}}\right)^{-4}$	$18^{\frac{2}{3}}$	$\frac{1}{(\sqrt[5]{21})^4}$
$21^{-\frac{4}{5}}$	$(\sqrt[3]{18})^2$	$(\sqrt{18})^3$	$\left(18^{\frac{1}{3}}\right)^2$

Record your matches below.

Equivalent Expression

1.6.5 Try It: Writing Equivalent Expressions

1. Complete the table.

$(a)^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^{b}$	$(a^{b})^{\frac{1}{c}}$	$(\sqrt[c]{a})^{b}$	$\sqrt[c]{a^{b}}$
$(100)^{\frac{3}{2}}$	$\left((100)^{\frac{1}{2}}\right)^{3}$			
			$(\sqrt[4]{60})^{5}$	
				$\sqrt[5]{8^{3}}$
		$((27)^{2})^{\frac{1}{3}}$		

2. Based on the equivalent expressions in the table, write three equivalent expressions for $(\sqrt{17})^{8}$.

1.6.6 Your Turn!

1. Rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$, where a, b, and c are integers.

$(\sqrt[3]{72})^{5}$

$\sqrt{5^{3}}$

$\dfrac{1}{(\sqrt[3]{20})^{4}}$

$\dfrac{1}{\sqrt{6}}$

1.6.7 Wrap-Up: Cool Down

1. Circle the expressions equivalent to $(25^3)^{\frac{1}{2}}$.

$(25)^{\frac{3}{2}}$ $\quad$ $\left(\sqrt{25}\right)^3$ $\quad$ $(25)^{\frac{2}{3}}$ $\quad$ $(25^{\frac{1}{3}})^2$ $\quad$ $\sqrt{25^3}$

Unit 1, Lesson 7: Evaluating Exponential Expressions

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can evaluate numerical expressions involving rational exponents.
- ☐ I can generate equivalent numerical expressions involving rational exponents.